Section 21.3 Notation for functions

We have already seen that a function can be written in any of these ways:

(i) $f(x) = 3x - 2$ (ii) $f: x \to 3x - 2$ (iii) $y = 3x - 2$

In each case, the output is $(3x - 2)$ when the input is x.

The notation $\boldsymbol{f(3)}$ is used to represent the output number when 3 is the input number.
If $f(x) = 3x - 2$
then $f(3) = 3(3) - 2 = 9 - 2 = 7$.

While $f(x)$ is generally used to describe a function, other letters such as $g(x)$ and $h(x)$ are used when we are dealing with more than one function.

Example 1

The functions f and g are defined as follows

$f: x \to x + 5$ and $g: x \to x^2 - 1$.

Find (i) $f(3)$ (ii) $f(-4)$ (iii) $3f(-2)$ (iv) $g(4)$ (v) $g(-3)$

(i) $f(x) = x + 5$
$f(3) = 3 + 5 = 8$

(ii) $f(x) = x + 5$
$f(-4) = -4 + 5 = 1$

(iii) $f(x) = x + 5$
$3f(-2) = 3(-2 + 5)$
$= 3(3) = 9$

(iv) $g(x) = x^2 - 1$
$g(4) = 4^2 - 1 = 16 - 1 = 15$

(v) $g(x) = x^2 - 1$
$g(-3) = (-3)^2 - 1 = 9 - 1 = 8$

Example 2

$f(x) = 3x - 4$ defines a function.

(i) If $f(x) = 5$, find the value of x.
(ii) If $f(a) = 11$, find the value of a.

(i)
$f(x) = 3x - 4$
$\therefore\ 3x - 4 = 5$
$3x = 9$
$x = 3$

(ii)
$f(a) = 3a - 4$
$\therefore\ 3a - 4 = 11$
$3a = 15$
$a = 5$

Exercise 21.3

1. If $f(x) = 3x$, find

(i) $f(1)$ (ii) $f(3)$ (iii) $f(4)$ (iv) $f(6)$ (v) $f(-2)$

2. If $f(x) = 2x + 3$, find

(i) $f(2)$ (ii) $f(3)$ (iii) $f(5)$ (iv) $f(0)$ (v) $f(-3)$

3. If $f(x) = 3x - 1$, find

(i) $f(1)$ (ii) $f(2)$ (iii) $f(4)$ (iv) $f(5)$ (v) $f(-1)$

4. If $f(x) = x^2 - 3$, find

(i) $f(0)$ (ii) $f(1)$ (iii) $f(2)$ (iv) $f(-2)$ (v) $f(-4)$

5. If $f(x) = 5 - 2x$, find

(i) $f(0)$ (ii) $f(2)$ (iii) $f(-3)$ (iv) $f(-\frac{1}{2})$ (v) $f(k)$

6. If $f(x) = 5x - 2$, solve the following equations:

(i) $f(x) = 8$ (ii) $f(x) = 3$ (iii) $f(k) = -12$

7. If $f(x) = 2x - 5$, find

(i) $f(3)$ (ii) $f(0)$ (iii) $f(1) + f(4)$ (iv) $2f(-3)$ (v) $f(3) + f(-3)$

8. If $f(x) = 3x - 2$ and $g(x) = 2 - 4x$, solve the following equations:

(i) $f(x) = 4$ (ii) $g(x) = -10$ (iii) $g(x) = f(4)$

9. If $f(x) = 5x - 2$ and $x \in \{1, 2, 3, 4\}$, write down the four couples of the function.

10. A function is defined by $f: x \to x^2 + 3x - 2$. Find

(i) $f(1)$ (ii) $f(2)$ (iii) $f(3)$ (iv) $f(0)$ (v) $f(-2)$

11. A function is defined by $f(x) = 5x - 4$.
If the domain of f is $\{0, 1, 2, 3, 4\}$, find the range of f.

12. $g: x \to 3x + 4$ is a function.

(i) Find $g(4)$. (ii) If $g(x) = 10$, find x. (iii) If $g(k) = -2$, find k.

13. If $h(x) = 3x - 5$ which *one* of the following is not a couple of the function?

(i) $(2, 1)$ (ii) $(4, 7)$ (iii) $(2, -1)$ (iv) $(0, -5)$

14. The domain of the function $f: x \to x^2 - 3x + 4$ is $\{-2, -1, 0, 1, 2)$.

(i) List the five couples of the function.

(ii) Write down the range of the function.